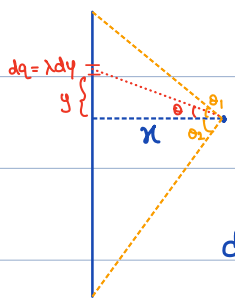
Electrostatics:

- $\vec{F} = \frac{1}{4\pi\varepsilon_0}\frac{qQ}{R^2}\hat{e}_r = Q\vec{E}$ $\quad \varepsilon_0 = 8.85\times10^{-12}\ C^2N^{-1}m^{-2}$

- interaction between any two charges is completely unaffected by the presence of others

- $\vec{F} = \vec{F_1}+\vec{F_2}+\cdots = Q\left[\frac{1}{4\pi\varepsilon_0}\frac{q_1}{R_1^2}\hat{e}_{r_1} + \frac{1}{4\pi\varepsilon_0}\frac{q_2}{R_2^2}\hat{e}_{r_2}+\cdots\right] = Q\vec{E}$

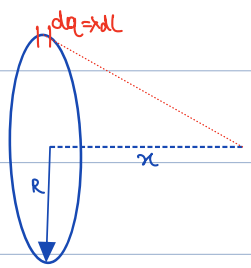
- 

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{\lambda dy}{x^2+y^2} = \frac{1}{4\pi\varepsilon_0}\frac{\lambda\cdot x\cdot\sec^2\theta\, d\theta}{x^2\sec^2\theta} = \frac{\lambda}{4\pi\varepsilon_0 x}d\theta$$

$$\tan\theta = \frac{y}{x}$$

$$dE_x = \frac{\lambda}{4\pi\varepsilon_0 x}\cos\theta\, d\theta \Rightarrow E_x = \frac{\lambda}{4\pi\varepsilon_0 x}(\sin\theta_1+\sin\theta_2)\hat{x}$$

$$dE_y = \frac{\lambda}{4\pi\varepsilon_0 x}\sin\theta\, d\theta \Rightarrow E_y = \frac{\lambda}{4\pi\varepsilon_0 x}(\cos\theta_1-\cos\theta_2)\hat{y}$$

- 

$$dE = \frac{1}{4\pi\varepsilon_0}\frac{\lambda dL}{R^2+x^2}$$

$$dE_x = \frac{1}{4\pi\varepsilon_0}\frac{\lambda dL}{R^2+x^2}\cdot\frac{x}{(R^2+x^2)^{1/2}} \Rightarrow E_x = \frac{\lambda}{2\varepsilon_0}\cdot\frac{Rx}{(R^2+x^2)^{3/2}}\hat{x}$$

$$dE_y = 0 \Rightarrow E_y = 0$$

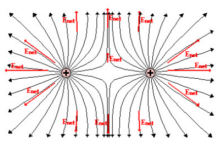
- For circular disc, we take multiple rings

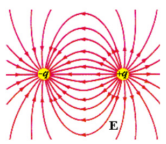
$$dE_x = \frac{\sigma dr\cdot r x}{2\varepsilon_0(r^2+x^2)^{3/2}} \Rightarrow E_x = \frac{\sigma}{2\varepsilon_0}\cdot\left(1-\frac{x}{(x^2+R^2)^{1/2}}\right)\hat{x} \quad E_y = 0$$

- Electrostatic field is conservative (all central forces are conservative)
- $\vec{E}(\vec{r}) = -\vec{\nabla}(\phi(\vec{r})) \iff \vec{\nabla} \times \vec{E}(\vec{r}) = 0$
- $\phi(\vec{r}) = \frac{1}{4\pi\varepsilon_0} \frac{q}{|\vec{r} - \vec{r}'|} + C$

- Line charge: (λ) $\vec{E} = \frac{1}{4\pi\varepsilon_0} \int \frac{\lambda \, dl'}{|\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}')$
- Surface charge: (σ) $\vec{E} = \frac{1}{4\pi\varepsilon_0} \int \frac{\sigma \, da'}{|\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}')$
- Volume charge: (ρ) $\vec{E} = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho \, dv'}{|\vec{r} - \vec{r}'|^3} (\vec{r} - \vec{r}')$

- Field lines: Never cross each other.





Field lines that originate from a +ve charge must either pass through a surface or terminate at a negative charge. A charge outside the surface won't contribute to the flux since its field lines pass in one side and come out through the other.

- Flux: $\int_S \vec{F} \cdot d\vec{a}$
- For any closed surface: $\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{enc}}{\varepsilon_0}$
- Gauss law integral form: $\vec{\nabla} \cdot \vec{E} = \rho/\varepsilon_0$
- spherical symmetry → concentric sphere
  cylindrical symmetry → coaxial cylinder
  plane symmetry → pill box

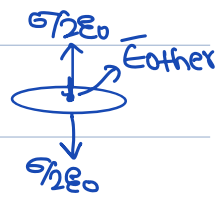
- Work done in assembling: $W = \frac{1}{2}\int \phi \rho \, dV$
- $\bar{E} = -\nabla V$; $\bar{\nabla} \times \bar{E} = 0 \Rightarrow \frac{dE_x}{dy} = \frac{dE_y}{dx}$; $\frac{dE_z}{dx} = \frac{dE_x}{dz}$; $\frac{dE_y}{dz} = \frac{dE_z}{dy}$
- $\bar{\nabla} \cdot \bar{E} = \frac{\rho}{\varepsilon_0} = -\nabla^2 V$

  $\Rightarrow \nabla^2 V = -\rho/\varepsilon_0$ poisson's equation

  $\Rightarrow \nabla^2 V = 0$ in charge free region laplace equation
- $E_2^{\perp} - E_1^{\perp} = \frac{\sigma}{\varepsilon_0}$; $E_2^{\parallel} = E_1^{\parallel}$; $V_2 = V_1$ (Boundary cond^{n}s)
- $W = \frac{\varepsilon_0}{2} \int_{\text{all space}} |\bar{E}|^2 dV$

- Energy density stored in field $= \frac{\varepsilon_0}{2}|\bar{E}|^2$
- Energy density stored in charges $= \frac{\rho\phi}{2}$
- work, energy → can't superimpose
- Conductors: $E = 0$ inside conductor. If we keep conductor in field, induced charges develop an and an opposite field is generated opposite $\rho = 0$ inside conductor; any net charge resides on the surface; A conductor is an equipotential surface; E is $\perp$ to surface just outside conductor.
- In presence of electric field, a surface charge

experiences force per unit area $\bar{f} = \sigma \bar{E}$



$\bar{E} = \bar{E}_{patch} + \bar{E}_{other}$

$\bar{E}_{above} = \bar{E}_{other} + \sigma/2\varepsilon_0 \quad \bar{E}_{below} = \bar{E}_{other} - \frac{\sigma}{2\varepsilon_0}$

$\Rightarrow \bar{E}_{other} = \frac{\bar{E}_{above} + \bar{E}_{below}}{2} = \frac{\sigma}{2\varepsilon_0} \Rightarrow \bar{f} = \sigma \bar{E} = \frac{\sigma^2}{2\varepsilon_0}$

outwards electrostatic pressure on surface

- $C = Q/V$. measure $\bar{E}$, then $\bar{V}$, then Q/V

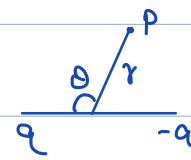
parallel plate: $|\bar{E}| = \sigma/\varepsilon_0 = Q/\varepsilon_0 A$; $V = Qd/A\varepsilon_0 \rightarrow C = \varepsilon_0 A/d$

spherical: $E = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{r}$; $V = -\int_a^b \bar{E}\cdot d\bar{r}$; $C = 4\pi\varepsilon_0 \frac{ab}{b-a}$

cylindrical: $E = \frac{Q}{2\pi\varepsilon_0 rL}$; $V = -\int_a^b \bar{E}\cdot d\bar{r}$; $C = 2\pi\varepsilon_0 L/\ln(b/a)$

- $W_{capacitor} = \frac{1}{2}CV^2$

- $V_{dipole} = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2}$



$E_{dipole} = -\frac{dV}{dr} - \frac{1}{r}\frac{dV}{d\theta} = \frac{p}{4\pi\varepsilon_0 r^3}[2\cos\theta\,\hat{e}_r + \sin\theta\,\hat{e}_\theta]$

$= \frac{1}{4\pi\varepsilon_0 r^3}[3(\bar{p}\cdot\hat{r})\hat{r} - \bar{p}]$

- 1D laplace equation:

cartesian: $\frac{d^2V}{dx^2} = 0 \Rightarrow V = mx + b$

→ $V(x)$ is average of $V(x+a)$, $V(x-a)$ for any a

→ solutions of laplace's eqn can't have local maxima or minima

- spherical: $\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right)=0 \Rightarrow V=-\frac{a}{r}+b$

  cylindrical: $\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial V}{\partial r}\right)=0 \Rightarrow V=a\ln(r)+b$

- multipole expansion:

  monopole: $\sum_{i=1}^{n} q_i = Q$; $V_m(\vec{r})=\frac{Q}{4\pi\varepsilon_0 r}$

  dipole: $\vec{p}=\sum_{i=1}^{n} q_i\vec{r}_i$; $V_d(\vec{r})=\frac{\vec{p}\cdot\hat{r}}{4\pi\varepsilon_0 r^2}$

  quadrupole: $Q_{ij}=\sum_k q_k(3r_{ki}r_{kj}-r_k^2\delta_{ij})$

  $Q_{xx}+Q_{yy}+Q_{zz}=0$

  $Q_{jk}=Q_{kj}\quad (j\neq k)$

  $$V_{Quad}=\frac{1}{4\pi\varepsilon_0 R^3}\frac{1}{2R^2}\begin{bmatrix}x & y & z\end{bmatrix}\begin{bmatrix}Q_{xx} & Q_{xy} & Q_{xz}\\ Q_{yx} & Q_{yy} & Q_{yz}\\ Q_{zx} & Q_{zy} & Q_{zz}\end{bmatrix}\begin{bmatrix}x\\ y\\ z\end{bmatrix}$$

- Change of origin: monopole moment is unique property ; $\vec{p}_{new}=\vec{p}-Q\vec{a}$ ($\vec{a}\rightarrow$ changed origin)
- $\vec{p}=\alpha\vec{E}$ $\alpha\rightarrow$ atomic polarizability
- $\vec{\tau}=\vec{p}\times\vec{E}$
- non-uniform field $\Rightarrow \vec{F}=(\vec{p}\cdot\nabla)\vec{E}$

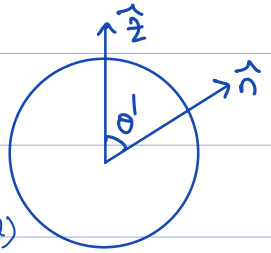
$\nabla'\left(\frac{1}{R}\right)=\frac{\hat{R}}{R^2}$

- Polarization $\vec{P}=$ dipole moment $\vec{p}$ per unit volume
- Potential due to an polarized object

$$V=\frac{1}{4\pi\varepsilon_0}\oint\frac{\sigma_b}{R}da'+\frac{1}{4\pi\varepsilon_0}\int_V\frac{\rho_b}{R}d\tau' \qquad \sigma_b=\vec{P}\cdot\hat{n} \qquad \rho_b=-\nabla\cdot\vec{P}$$

- uniformly polarized $\Rightarrow \rho_b=0$
- $\nabla\cdot\vec{D}=\rho_f$; $\oint\vec{D}\cdot d\vec{a}=(Q_f)_{enc}$; $\vec{D}=\varepsilon_0\vec{E}+\vec{P}$

- linear: $\bar{P} = \varepsilon_0 \chi_e \bar{E} \Rightarrow \bar{D} = \varepsilon_0 (1+\chi_e)\bar{E} = \varepsilon_0 \varepsilon_r \bar{E} = \varepsilon \bar{E}$
  (χ_e: dielectric susceptibility; ε_r: dielectric constant)
- Total work done $= \frac{1}{2}\int_{\text{all space}} \bar{D}.\bar{E}\, d\tau$
- No coulomb's law for $\bar{D}$; $\bar{\nabla}\times\bar{D} = \bar{\nabla}\times\bar{P}$
- $D^{\perp}_{above} - D^{\perp}_{below} = \sigma_f \qquad D^{\parallel}_{above} - D^{\parallel}_{below} = P^{\parallel}_{above} - P^{\parallel}_{below}$
- uniformly polarized sphere



$\vec{P} = P\hat{z} \Rightarrow \sigma_b = P\cos\theta' \; ; \; \rho_b = 0 \; ; \; R = (z^2+a^2-2az\cos\theta')^{1/2}$

$\phi = \frac{1}{4\pi\varepsilon_0}\int \frac{\sigma_b}{R}da' = \frac{1}{4\pi\varepsilon_0}\int_0^{2\pi}\int_0^{\pi} \frac{P\cos\theta' . a^2 \sin\theta' d\theta' d\phi'}{(z^2+a^2-2az\cos\theta')^{1/2}}$ (let $\cos\theta' = \mu$)

$\Rightarrow \phi = \frac{P}{6\varepsilon_0 z^2}\left[(z^2+a^2)(|z+a|-|z-a|) - 2a(|z+a|+|z-a|)\right]$

$z > a \Rightarrow \phi = \frac{Pa^3}{3\varepsilon_0 z^2} \Rightarrow \bar{E} = \frac{2\bar{P}a^3}{3\varepsilon_0 z^2} \qquad z < a \Rightarrow \phi = \frac{Pz}{3\varepsilon_0} \Rightarrow \bar{E} = \frac{-\bar{P}}{3\varepsilon_0}$

- homogeneous linear dielectric: $\rho_b = \left(\frac{-\chi_e}{1+\chi_e}\right)\rho_f$ i.e $\rho_b \propto \rho_f$
- energy of an ideal dipole $U = -\bar{p}.\bar{E} = \frac{-p^2}{4\pi\varepsilon_0 r^3}[3\cos^2\theta - 1]$
- interaction energy between 2 dipoles:



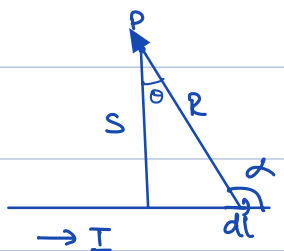
$U = -\bar{p}_1.\bar{E}_2 = -p_1 . \frac{1}{4\pi\varepsilon_0 r^3}[3(\bar{p}_2.\hat{r})\hat{r} - \bar{p}_2]$

$= \frac{\bar{p}_1.\bar{p}_2 - 3(\bar{p}_1.\hat{r})(\bar{p}_2.\hat{r})}{4\pi\varepsilon_0 r^3} = -\bar{p}_2\bar{E}_1$

Magnetostatics:

- $\vec{F}_{mag} = q(\bar{v} \times \bar{B})$ Lorentz force law
- $\vec{F} = q(\bar{E} + \bar{v} \times \bar{B})$ Lorentz force
- motion of charged particle is magnetic field is circular:

  $qvB = \frac{mv^2}{r} \Rightarrow p = qBr$ (cyclotron formula)
- magnetic forces do no work, can just alter the direction
- $\bar{F} = \int (\bar{K} \times \bar{B})\, da$ $\quad \bar{K} = \frac{d\bar{I}}{dl_\perp} = \sigma \bar{v}$; $\bar{F} = \int (\bar{J} \times \bar{B})\, d\tau$ $\quad \bar{J} = \frac{d\bar{I}}{da_\perp} = \rho \bar{v}$
- $\bar{\nabla} \cdot \bar{J} = -\frac{d\rho}{dt}$ continuity equation
- steady currents $\Rightarrow \bar{\nabla} \cdot \bar{J} = 0$
- $\bar{B}(\bar{r}) = \frac{\mu_0}{4\pi} \int \frac{I d\bar{l} \times \hat{R}}{R^2}$; $\mu_0 = 4\pi \times 10^{-7}\ N A^{-2}$ permeability of free space
- $l = -s \cot\alpha \Rightarrow dl = s \operatorname{cosec}^2\alpha\, d\alpha$



$$B = \frac{\mu_0 I}{4\pi} \int \frac{s \operatorname{cosec}^2\alpha\, d\alpha \sin\alpha}{s^2 \operatorname{cosec}^2\alpha} = \frac{\mu_0 I}{4\pi s} [\sin\theta_1 - \sin\theta_2]$$

Infinitely long wire: $B = \frac{\mu_0 i}{2\pi s}$

- Ring: $\frac{\mu_0 I a^2}{2(a^2+z^2)^{3/2}}$
- Solenoid: $\mu_0 n I$ ($n \rightarrow$ number of turns per unit length);

no radial, circumferential component $\quad \frac{\mu_0 n I}{2}(\cos\alpha_1 + \cos\alpha_2)$; outside is 0

- $\oint \bar{B} \cdot d\bar{l} = \mu_0 I_{enc}$ Ampere's law integral form
- $\bar{\nabla} \times \bar{B} = \mu_0 \bar{J}$ Ampere's law differential form
- $\bar{\nabla} \cdot \bar{B} = 0 \rightarrow$ magnetic field is solenoidal

$\nabla \cdot \hat{R}/R^2 = 4\pi \delta^3(\bar{R})$

- over a sheet: $B = \frac{\mu_0 K}{2}$
- maxwell's equations:

$\nabla \cdot \bar{E} = \rho/\varepsilon_0 \quad \nabla \times \bar{E} = 0 \quad \nabla \cdot \bar{B} = 0 \quad \nabla \times \bar{B} = \mu_0 \bar{J}$

- magnetic fields are produced by electric charges in motion. and it takes another moving charge to feel it.
- $\bar{B}(\bar{r}) = \nabla \times \bar{A}(\bar{r})$ $\bar{A}(\bar{r}) \rightarrow$ magnetic vector potential
- $A = \frac{\mu_0}{4\pi} \int \frac{I d\bar{l}}{r} = \frac{\mu_0}{4\pi} \int \frac{\bar{J} d\tau}{r}$

$\int \nabla^2 (1/r)\, d\tau = -4\pi$

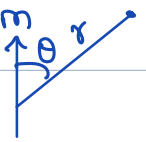
- $\nabla^2 A = -\mu_0 J$
- $\int \bar{B} \cdot d\bar{a} = \oint \bar{A} \cdot d\bar{l} = \phi$ (flux)
- $\nabla(\nabla \cdot \bar{A}) = 0$ If $\bar{A} = \bar{A}_0 + \nabla\lambda$ such that $\nabla \cdot \bar{A} = 0$ ($\lambda \rightarrow$ scalar)

$\Rightarrow \nabla^2 \lambda = -\nabla \cdot \bar{A}_0$ (non-uniqueness of vector potential)

- $B^{\perp}_{above} = B^{\perp}_{below}$ $\quad B^{\parallel}_{above} - B^{\parallel}_{below} = \mu_0 K$ ($\perp$ to current)

$\vec{A}_{above} = \vec{A}_{below}$

- magnetic monopole term vanishes always



- $\vec{A}_{dip}(\bar{r}) = \frac{\mu_0}{4\pi} \frac{\bar{m} \times \hat{r}}{r^2}$ $\quad \bar{m} = I \int d\vec{a}$
- $\bar{B}_{dip}(\bar{r}) = \frac{\mu_0 m}{4\pi r^3} (2\cos\theta\, \hat{e}_r + \sin\theta\, \hat{e}_\theta) = \frac{\mu_0}{4\pi r^3} [3(\bar{m} \cdot \hat{r})\hat{r} - \bar{m}]$
- Rotating disc: $dq = \sigma \cdot 2\pi r\, dr$ $\quad dI = \frac{dq}{T} = \frac{\omega\, dq}{2\pi} = \sigma \omega r\, dr$

$dm = dI \cdot \pi r^2 = \pi \sigma \omega r^3 dr \Rightarrow \vec{m} = \pi \sigma \vec{\omega} R^4/4$

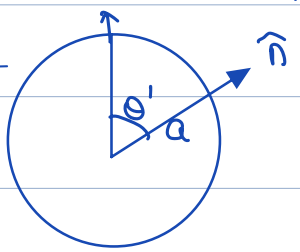
- diamagnetic: not permanent magnetic moment, develop magnetization opposite to $\bar{B}$, change in orbital speed of electron, always present
- paramagnet: permanent magnetic moment, develop magnetization parallel to $\bar{B}$, spins of unpaired electrons experiencing a torque
- Ferromagnetism: domains align together
- $\bar{\tau} = \bar{m} \times \bar{B}$ (responsible for paramagnetism)
- $i = \frac{ev}{2\pi R} \Rightarrow \bar{m} = -\frac{1}{2} ev R \hat{z}$

  before: $\frac{e^2}{4\pi\varepsilon_0 R^2} = \frac{m_e v^2}{R}$ after: $\frac{e^2}{4\pi\varepsilon_0 R^2} + ev\bar{B} = \frac{m_e \bar{v}^2}{R}$

  $\Rightarrow \bar{v} > v$ $\Delta v = \frac{eRB}{2m_e} \Rightarrow \Delta\vec{m} = -\frac{e^2R^2}{4m_e}\vec{B}$ (diamagnet-ism)

  change in magnetic field produces an electric field which does the work
- magnetization $\bar{M} = \bar{m}/\tau$
- $\bar{A}(\bar{r}) = \frac{\mu_0}{4\pi}\int \frac{(\bar{\nabla}\times\bar{m})}{R} d\tau + \frac{\mu_0}{4\pi}\oint_S \frac{(\bar{M}\times\hat{n})}{R} da$ $\bar{J}_b = \bar{\nabla}\times\bar{m}$, $\bar{K}_b = \bar{M}\times\hat{n}$
- $\bar{\nabla}\cdot\bar{J}_b = 0$ uniformly magnetized $\Rightarrow J_b = 0$
- $\bar{H} = \frac{\bar{B}}{\mu_0} - \bar{M}$ $\bar{\nabla}\times\bar{H} = \bar{J}_f$ $\oint \bar{H}\cdot d\bar{l} = (I_f)_{enc}$
- $\bar{\nabla}\cdot\bar{H} = -\bar{\nabla}\cdot\bar{M}$

- $H^{\perp}_{above} - H^{\perp}_{below} = M^{\perp}_{below} - M^{\perp}_{above}$ $\quad H^{\parallel}_{above} - H^{\parallel}_{below} = \vec{K}_f \times \hat{n}$
- linear: $\vec{M} = \chi_m \vec{H} \Rightarrow B = \mu H$; $\mu = \mu_0(1+\chi_m)$; $\mu_r = 1+\chi_m$
  (magnetic susceptibility; permeability; relative permeability)
- uniformly magnetized sphere



$\bar{M} = M\hat{z}$; $J_b = 0$, $\bar{K}_b = M\sin\theta' \hat{e}_\phi$ $\quad R = (z^2 + a^2 - 2az\cos\theta')^{1/2}$

$\bar{A}(\bar{r}) = \frac{\mu_0}{4\pi} \oint \frac{M\sin\theta' \hat{e}_\phi \cdot a^2 \sin\theta' \, d\theta' d\phi'}{(z^2+a^2-2az\cos\theta')^{1/2}} \Rightarrow \bar{B}_{inside} = \frac{2}{3}\mu_0 \bar{M}$.

Outside it's similar to pure dipole of $\bar{m} = \frac{4}{3}\pi a^3 \bar{M}$

An electron of mass m and charge $-q$ is moving under the influence of a uniform electric field, $\bar{E} = E_0\hat{y}$ and a uniform magnetic field, $\bar{B} = B_0\hat{z}$. The initial position of the electron is the origin (0, 0,0) and the initial velocity is $\bar{v} = v_0\hat{x}$.
(i) Write the equations of motion. (ii) Determine the trajectory $\bar{r}(t)$ of the electron at time $t > 0$.

Let $v = v_x\hat{i} + v_y\hat{j} + v_z\hat{k}$ $\quad v_z = 0 = a_z$

$a_x = -\frac{qB_0 v_y}{m}$ $\quad a_y = \frac{qB_0 v_x - qE_0}{m}$ $\quad \frac{d^2v_x}{dt^2} = -\frac{qB_0}{m}\left[\frac{qB_0 v_x - qE_0}{m}\right]$

$\frac{dv_x}{dt} = -\frac{qB_0 v_y}{m}$ $\quad \frac{dv_y}{dt} = \frac{qB_0 v_x - qE_0}{m}$ $\Rightarrow \frac{d^2v_x}{dt^2} = -a^2 v_x + ab$

$\Rightarrow v_x = A\sin(at+\delta) + \frac{b}{a}$ $\quad \cdot A = v_0 - E_0/B_0$ $\quad \frac{qB_0}{m} = q$ $\quad \frac{qE_0}{m} = b$

$\Rightarrow v_y = -A\cos(at+\frac{\pi}{2})$

$$v_{\bar{E}\times\bar{B}} = (v_0 - E_0/B_0)\cos\left(\frac{qB_0}{m}t\right) + E_0/B_0 \qquad v_{\bar{E}} = -(v_0 + E_0/B_0)\sin\left(\frac{qB_0}{m}t\right)$$

$x(t) = \frac{v_0 - E_0/B_0}{qB_0/m}\sin\left(\frac{qB_0}{m}t\right) + \frac{E_0 t}{B_0}$ $\quad (\bar{E}\times\bar{B})$

$y(t) = \frac{v_0 - E_0/B_0}{qB_0/m}\cos\left(\frac{qB_0}{m}t\right) - \frac{v_0 - E_0/B_0}{qB_0/m}$ $\quad (\bar{E})$

Electrodynamics:

- $\vec{J} = \sigma \vec{f}$ $\quad \vec{f} \rightarrow$ force per unit charge $= \vec{E} + \vec{v} \times \vec{B}$
- $\sigma = 1/\rho$
- $\vec{J} = \sigma \vec{E}$ ($\vec{v}$ is small) Ohm's law

  $\sigma \rightarrow \infty$ for conductor
- $V = IR$ (other way of Ohm's law)
- $\vec{\nabla} \cdot \vec{E} = \frac{1}{\sigma} \vec{\nabla} \cdot \vec{J} = 0 \Rightarrow$ Laplace eqn holds within homogeneous ohmic material with steady current.
- Emf: work done per unit charge

  $\mathcal{E} = \oint \vec{f} \cdot d\vec{l} = \oint \vec{f}_s \cdot d\vec{l}$ $\quad f_s \rightarrow$ source
- motional emf : $\mathcal{E} = Bvh$
- $\mathcal{E} = -d\Phi/dt$
- $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ changing magnetic field induces an electric field (faraday's law)
- Lenz's law: The induced current will flow in such a direction that the flux it produces tends to cancel the change
- $\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi}{dt}$ Faraday's law in integral form

- $\bar{\nabla}\cdot\bar{E}=0$, $\bar{\nabla}\bar{B}=0$, $\bar{\nabla}\times\bar{B}=\mu_0 J$
- $\bar{\nabla}\times\bar{E}=-\frac{d}{dt}(\bar{\nabla}\times\bar{A})=-\nabla\times(\frac{d\bar{A}}{dt}) \Rightarrow \bar{E}=-\nabla\phi-\frac{d\bar{A}}{dt}$
- Flux due to first loop on second loop depends on current in first loop. The proportionality constant is mutual inductance M_{21}

  $\phi_2 = M_{21} I_1$; $M_{21}=\frac{\mu_0}{4\pi}\oint\oint \frac{d\bar{l}_1 . d\bar{l}_2}{R}$; $M_{12}=M_{21}$
- Since flux in a loop changes, it induces an emf in itself
- Solenoids : $B=\mu_0 n I \Rightarrow \phi = l n_2 B . a = \mu_0 n_1 n_2 \pi a^2 I l$ $= LI$
- $W=\frac{1}{2}LI^2=\frac{1}{2}\int_v (\bar{A}.\bar{J})\,d\tau=\frac{1}{2\mu_0}\int_{\text{all space}} B^2\,d\tau$
- $\bar{\nabla}\cdot\bar{E}=\rho/\varepsilon_0$ $\bar{\nabla}\cdot\bar{B}=0$ $\bar{\nabla}\times\bar{E}=-\frac{d\bar{B}}{dt}$ $\bar{\nabla}\times\bar{B}=\mu_0(\bar{J}+\varepsilon_0\frac{d\bar{E}}{dt})$

  A changing electric field induces a magnetic field
- $\varepsilon_0\frac{d\bar{E}}{dt}=\frac{d\bar{D}}{dt}=\bar{J}_d$ (displacement current)

  → current only in a sense that it produces a magnetic field, makes total current continuous across discontinuities in conduction current. J_d in good conductor is negligible

compared to conduction current at optical frequencies

- Maxwell's equations in matter:

$$\nabla\cdot\vec{D}=\rho_f \quad \nabla\times\vec{E}=-\frac{\partial\vec{B}}{\partial t} \quad \nabla\cdot\vec{B}=0 \quad \nabla\times\vec{H}=\vec{J}_f+\frac{\partial\vec{D}}{\partial t}$$

$$\oint_S \vec{D}\cdot d\vec{a}=Q_f \quad \oint_S \vec{B}\cdot d\vec{a}=0 \quad \oint_P \vec{E}\cdot d\vec{l}=-\frac{d}{dt}\int_S \vec{B}\cdot d\vec{a}$$

$$\oint_P \vec{H}\cdot d\vec{l}=I_f+\frac{d}{dt}\int \vec{D}\cdot d\vec{a}$$

- Boundary conditions:

$$D_1^{\perp}-D_2^{\perp}=\sigma_f \quad B_1^{\perp}=B_2^{\perp} \quad E_1^{\parallel}=E_2^{\parallel} \quad H_1^{\parallel}-H_2^{\parallel}=\vec{K}_f\times\hat{n}$$

- $u_{em}=\frac{1}{2}\int\left(\varepsilon_0 E^2+\frac{B^2}{\mu_0}\right)d\tau$
- $\vec{S}=\frac{1}{\mu_0}(\vec{E}\times\vec{B})$ = energy per unit time per unit area

  $\vec{S}\cdot d\vec{a}$ is energy flux or energy flux density
- $\frac{d}{dt}(u_{mech}+u_{em})+\nabla\cdot\vec{S}=0$
- $\nabla^2 E=\mu_0\varepsilon_0\frac{\partial^2 E}{\partial t^2}$ $\quad \nabla^2 B=\mu_0\varepsilon_0\frac{\partial^2 B}{\partial t^2}$
- $\vec{E}(\vec{r},t)=|E_0|\hat{n}\cos(\vec{k}\cdot\vec{r}-\omega t+\delta)$

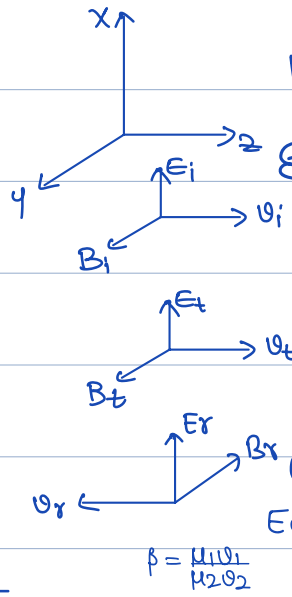
  $\vec{B}(\vec{r},t)=\frac{|E_0|}{c}(\hat{k}\times\hat{n})\cos(\vec{k}\cdot\vec{r}-\omega t+\delta)$

  $k \rightarrow$ gives direction of propogation and wavelength

  $\hat{n} \rightarrow$ polarization, direction of electric field
- $u_{em}=\varepsilon_0 E^2=\varepsilon_0 E_0^2\cos^2(\omega t+\vec{k}\cdot\vec{r}+\delta) \Rightarrow \langle u_{em}\rangle=\frac{1}{2}\varepsilon_0 E_0^2$
- $\vec{S}=c u_{em} \Rightarrow \langle S\rangle=\frac{1}{2}\varepsilon_0 E_0^2 c\,\hat{k}=I$ (intensity)

- $\vec{g} = \vec{S}/c^2 \Rightarrow \langle g \rangle = \frac{1}{2c}\varepsilon_0 E_0^2 \hat{k}$ $\quad \downarrow \frac{1}{2} n \varepsilon_0 c E_0^2 \rightarrow$ any medium
- $P = I/c$ (absorber), $2I/c$ (reflector)
- plane em waves in matter: $v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{n} \Rightarrow n = \sqrt{\varepsilon_r}$
- Normal incidence:

$$E_I(z,t) = E_{0,I}\, e^{i(k_1 z - \omega t)}\hat{x}$$
$$B_I(z,t) = \frac{E_{0,I}}{v_1} e^{i(k_1 z - \omega t)}\hat{y}$$
$$E_R(z,t) = E_{0,R}\, e^{i(-k_1 z - \omega t)}\hat{x}$$
$$B_R(z,t) = \frac{E_{0,R}}{v_1} e^{i(-k_1 z - \omega t)}(-\hat{y})$$
$$E_T(z,t) = E_{0,T}\, e^{i(k_2 z - \omega t)}\hat{x}$$
$$B_T(z,t) = \frac{E_{0,T}}{v_2} e^{i(k_2 z - \omega t)}\hat{y}$$



$$k_1 v_1 = k_2 v_2$$
$$\varepsilon_1 E_1^{\perp} = \varepsilon_2 E_2^{\perp}$$
$$E_1^{\parallel} = E_2^{\parallel}$$
$$B_1^{\perp} = B_2^{\perp}$$
$$\frac{B_1^{\parallel}}{\mu_1} = \frac{B_2^{\parallel}}{\mu_2}$$
$$E_{0,R} + E_{0,I} = E_{0,T}$$
$$E_{0,I} - E_{0,R} = \beta E_{0,T}$$
$$\beta = \frac{\mu_1 v_1}{\mu_2 v_2}$$

$$\frac{E_{0,R}}{E_{0,I}} = \frac{1-\beta}{1+\beta} = \frac{v_2 - v_1}{v_2 + v_1} = \frac{n_1 - n_2}{n_1 + n_2}$$

$$\frac{E_{0,T}}{E_{0,I}} = \frac{2\beta}{1+\beta} = \frac{2v_2}{v_1 + v_2} = \frac{2n_1}{n_1 + n_2}$$

reflection coefficient $R = \frac{I_R}{I_I} = \left(\frac{E_{0,R}}{E_{0,I}}\right)^2 = \left(\frac{n_1 - n_2}{n_1 + n_2}\right)^2$ $\qquad R + T = 1$

transmission coefficient $T = \frac{I_T}{I_I} = \frac{\varepsilon_2 v_2}{\varepsilon_1 v_1}\left(\frac{E_{0,T}}{E_{0,I}}\right)^2 = \frac{4 n_1 n_2}{(n_1 + n_2)^2}$

- $\boxed{\tilde{E}_{0_R} = \left(\frac{\alpha - \beta}{\alpha + \beta}\right)\tilde{E}_{0_I}, \quad \tilde{E}_{0_T} = \left(\frac{2}{\alpha + \beta}\right)\tilde{E}_{0_I}.}$; $\alpha \equiv \frac{\cos\theta_T}{\cos\theta_I}$; $\beta \equiv \frac{\mu_1 v_1}{\mu_2 v_2} = \frac{\mu_1 n_2}{\mu_2 n_1}$